One-Sample Tests of Hypothesis

Chapter 10

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Learning Objectives

LO10-1 Explain the process of testing a hypothesis

LO10-2 Apply the six-step procedure for testing a hypothesis

LO10-3 Distinguish between a one-tailed and a two-tailed test of hypothesis

LO10-4 Conduct a test of a hypothesis about a population mean

LO10-5 Compute and interpret a p-value

LO10-6 Use a t-statistic to test a hypothesis

LO10-7 Compute the probability of a Type II error

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Hypothesis Testing

- Hypothesis testing begins with a hypothesis statement about a population parameter

HYPOTHESIS A statement about a population parameter subject to verification

Examples

- The mean speed of automobiles passing milepost 150 on the West Virginia Turnpike is 68 mph
- The mean cost to remodel a kitchen is $20,000

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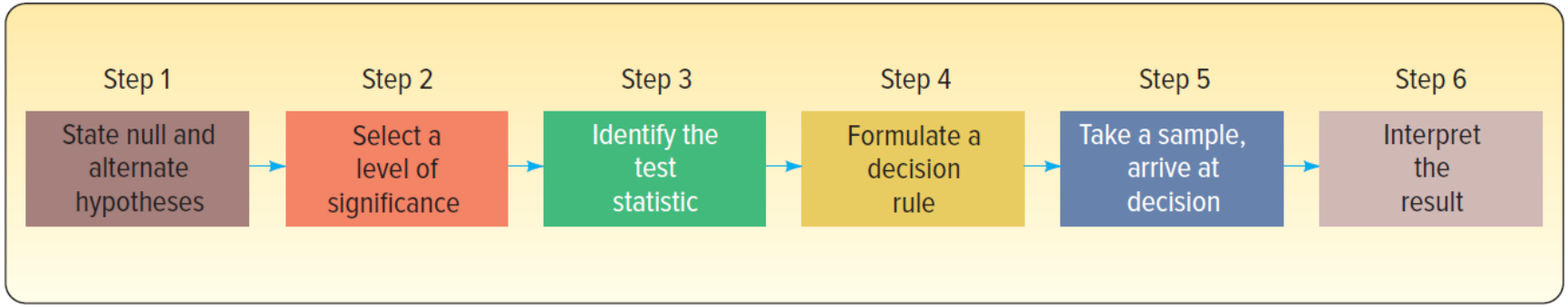
Hypothesis Testing (2 of 3)

- The objective of hypothesis testing is to verify the validity of a statement about a population parameter

HYPOTHESIS TESTING A procedure based on sample evidence and probability theory to determine whether the hypothesis is a reasonable statement.



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Step 1 of the Six-Step Process

- State the null hypothesis (H_0) and the alternate hypothesis (H_1)

> NULL HYPOTHESIS A statement about the value of a population parameter developed for the purpose of testing numerical evidence.

- The null hypothesis always includes the equal sign
  - For example; =, ≥, or ≤ will be used in H_0

> ALTERNATE HYPOTHESIS A statement that is accepted if the sample data provide sufficient evidence that the null hypothesis is false.

- The alternate hypothesis never includes the equal sign
  - For example; ≠, <, or > is used in H_1

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Step 2 of the Process

- Next, you select the level of significance,

> LEVEL OF SIGNIFICANCE The probability of rejecting the null hypothesis when it is true.

- Sometimes called the level of risk
- Can be any value between 0 and 1
- Traditionally,
  - .05 is used for consumer research projects
  - .01 for quality assurance
  - .10 for political polling

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Possible Error in Hypothesis Testing

- Since the researcher cannot study every item or individual in the population, error is possible

> TYPE I ERROR Rejecting the null hypothesis, H_0, when it is true.

- Type I error is designated with the Greek letter alpha, α

> TYPE II ERROR Not rejecting the null hypothesis when it is false.

- Type II error is designated with the Greek letter beta, β

	Researcher	
Null Hypothesis	**Does Not Reject H_0**	**Rejects H_0**
H_0 is true	Correct decision	Type I error
H_0 is false	Type II error	Correct decision

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Step 3 of the Process

- Then, select the test statistic

> TEST STATISTIC A value, determined from sample information, used to determine whether to reject the null hypothesis.

- In hypothesis testing for the mean, μ, when σ is known, the test statistic z is computed with the following formula

TESTING A MEAN, σ KNOWN

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \qquad (10\text{–}1)$$

- We can determine whether the distance between $\bar{x}$ and μ is statistically significant by finding the number of standard deviations $\bar{x}$ is from μ

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Step 4 of the Process

- Formulate the decision rule
- The decision rule is a statement of specific conditions under which the null hypothesis is rejected and the conditions under which it is not rejected
- The region or area of rejection defines the location of all the values that are either so large or so small that their probability of occurrence under a true null hypothesis is remote

CRITICAL VALUE The dividing point between the region where the null hypothesis is rejected and the region where it is not rejected.

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Critical Value

- The sampling distribution of the statistic z follows the normal distribution
- Here, an α of .05 is used in a one-tailed test
- The value 1.645 separates the regions where the null hypothesis is rejected and where it is not rejected
- The value 1.645 is the critical value

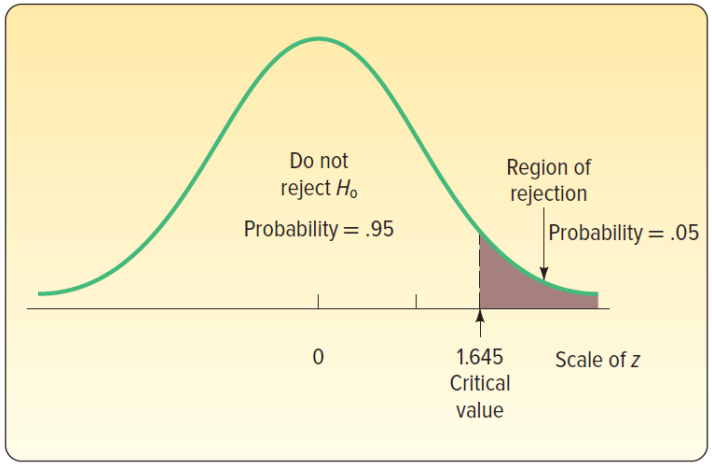

CHART 10–1 Sampling Distribution of the Statistic z, a Right-Tailed Test, .05 Level of Significance

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Steps 5 & 6 of the Six-Step Process

- Step 5 Make a decision
  - First, select a sample and compute the value of the test statistic
  - Compare the value of the test statistic to the critical value
  - Then, make the decision regarding the null hypothesis
- Step 6 Interpret the results
  - What can we say or report based on the results of the statistical test?

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Hypothesis Testing (3 of 3)

SUMMARY OF THE STEPS IN HYPOTHESIS TESTING

1. Establish the null hypothesis (H_0) and the alternate hypothesis (H_1).
2. Select the level of significance, that is, α.
3. Select an appropriate test statistic.
4. Formulate a decision rule based on steps 1, 2, and 3.
5. Make a decision regarding the null hypothesis based on the sample information.
6. Interpret the results of the test.

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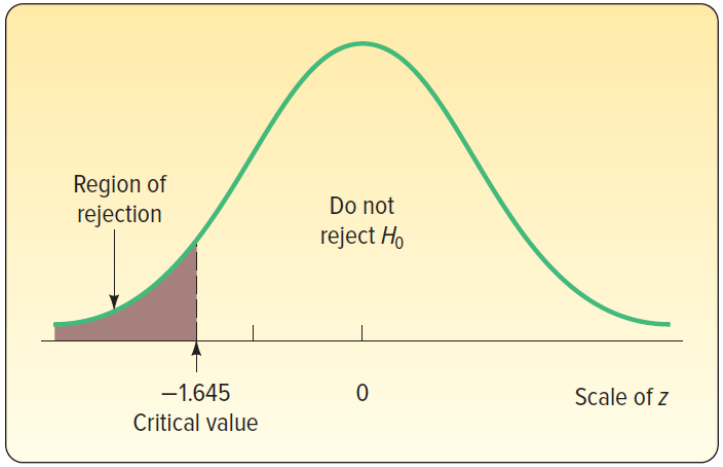
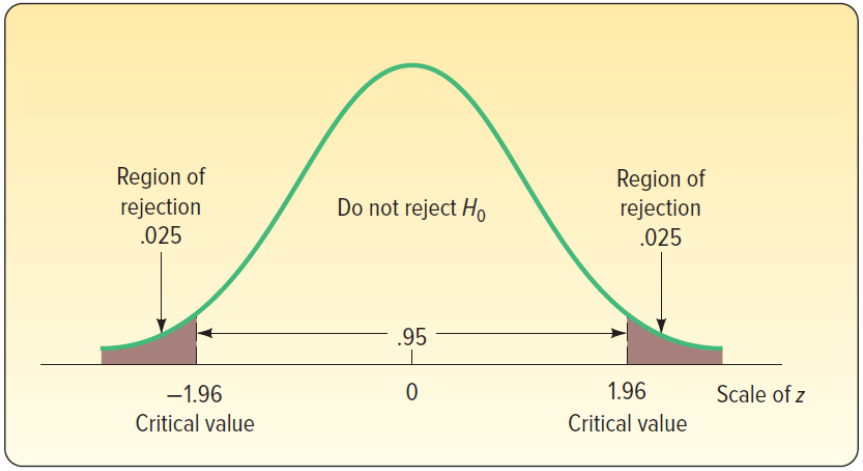
One-Tailed and Two-Tailed Tests

H_0: ≥ 60,000 miles
H_1: < 60,000 miles
with an $\alpha = .05$
Left-tailed test

H_0: = $65,000 per year
H_1: ≠ $65,000 per year
with an $\alpha = .05$
Two-tailed test





Note that the total area in the normal distribution is 1.0000.

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Two-Tailed Test Example, σ Known

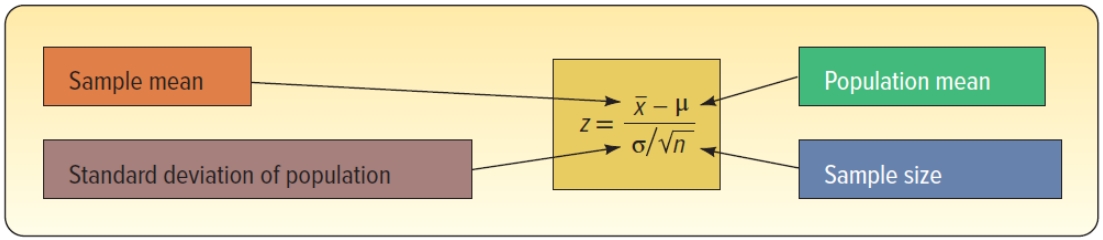
Jamestown Steel Company manufactures and assembles desks and other office equipment at several plants in New York State. At the Fredonia plant, the weekly production of the Model A325 desk follows a normal distribution with a mean of 200 and a standard deviation of 16. New production methods have been introduced and the vice president of manufacturing would like to investigate whether there has been a change in weekly production of the Model A325. Is the mean number of desks produced different from 200 at the .01 significance level?

Step 1: State the null hypothesis and alternate hypothesis.

$$H_0: \mu = 200 \text{ desks}$$
$$H_1: \mu \neq 200 \text{ desks}$$

Step 2: Select the level of significance. Here $\alpha = .01$

Step 3: Select the test statistic. In this example, we'll use z



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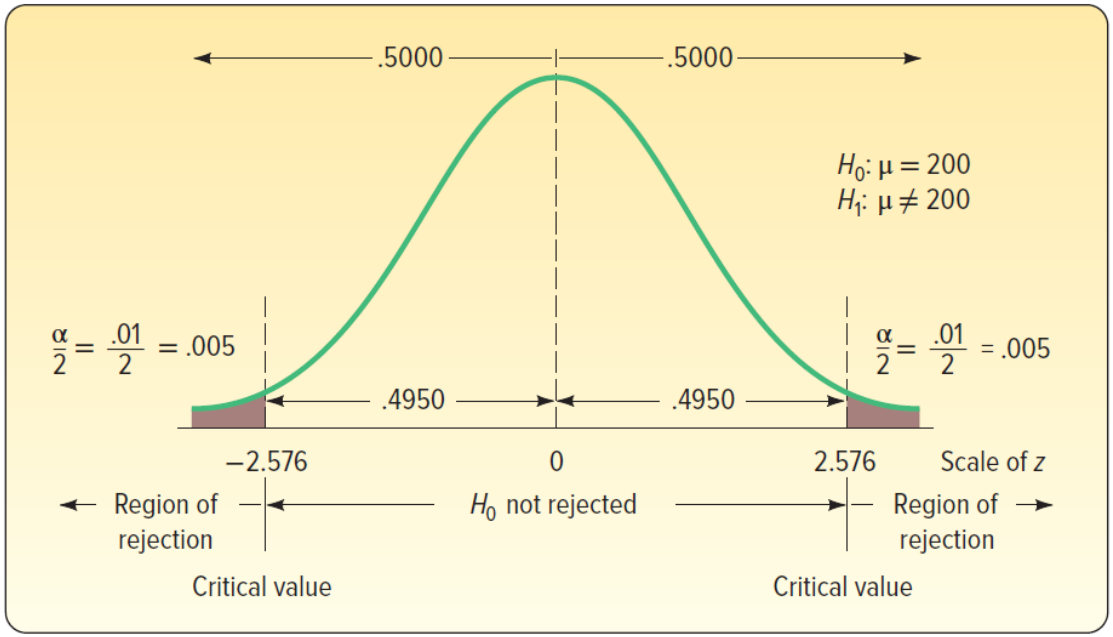
Two-Tailed Test Example, Known (2 of 3)

Step 4: Formulate the decision rule by first determining the critical values of z.



Decision Rule: If the computed value of z is not between −2.576 and 2.576, reject the null hypothesis. If z falls between −2.576 and 2.576, do not reject the null hypothesis.

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Two-Tailed Test Example, Known (3 of 3)

Step 5: Take sample, compute the test statistic, make decision.

The mean number of desks produced last year (50 weeks because the plant was shut down 2 weeks for vacation) is 203.5. The standard deviation of the population is 16 desks per week. Compute z with formula 10-1.

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{203.5 - 200}{16/\sqrt{50}} = 1.547$$

Decision: Because 1.547 does not fall in the rejection region, we decide not to reject H_0.

Step 6: Interpret the result.

We did not reject the null hypothesis, so we have failed to show that the population mean has changed from 200 per week.

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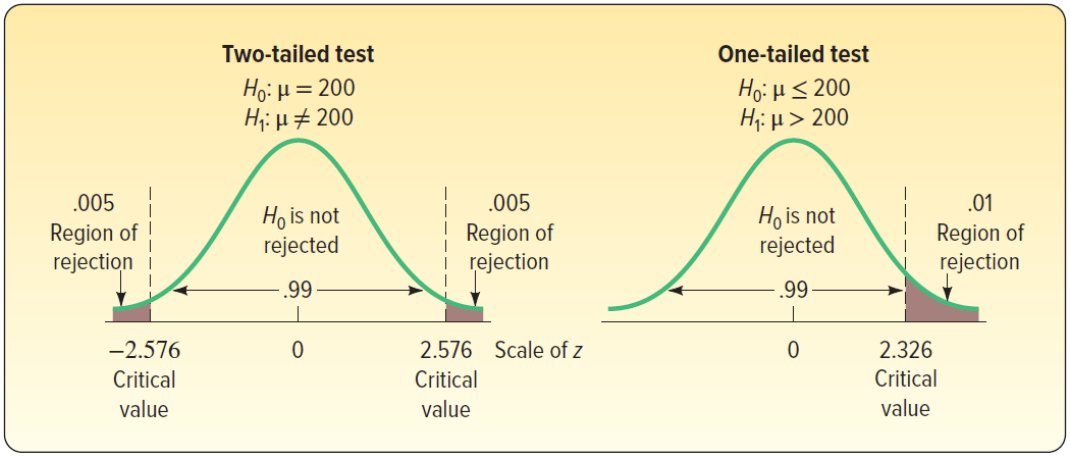
One-Tailed Test

Suppose instead of wanting to know if there had been a change in the mean number of desks assembled, the vice president wanted to know if there had been an increase in the number of units assembled. Can we conclude, because of the improved production methods, that the mean number of desks assembled in the last 50 weeks was more than 200? Use .

Before:
A two-tailed test
H_0: = 200 desks
H_1: ≠ 200 desks

Now:
A one-tailed test
H_0: ≤ 200 desks
H_1: > 200 desks



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The p-Value in Hypothesis Testing

p-VALUE The probability of observing a sample value as extreme as, or more extreme than the value observed, given that the null hypothesis is true.

- Compare the p-value with the level of significance, α
  - If the p-value is smaller than the significance level, reject H_0
  - If the p-value is larger than α, H_0 is not rejected
- A p-value not only results in a decision about H_0, but gives additional insight about the strength of that decision

INTERPRETING THE WEIGHT OF EVIDENCE AGAINST H_0

If the *p*-value is less than

(a) .10, we have *some* evidence that H_0 is not true.

(b) .05, we have *strong* evidence that H_0 is not true.

(c) .01, we have *very strong* evidence that H_0 is not true.

(d) .001, we have *extremely strong* evidence that H_0 is not true.

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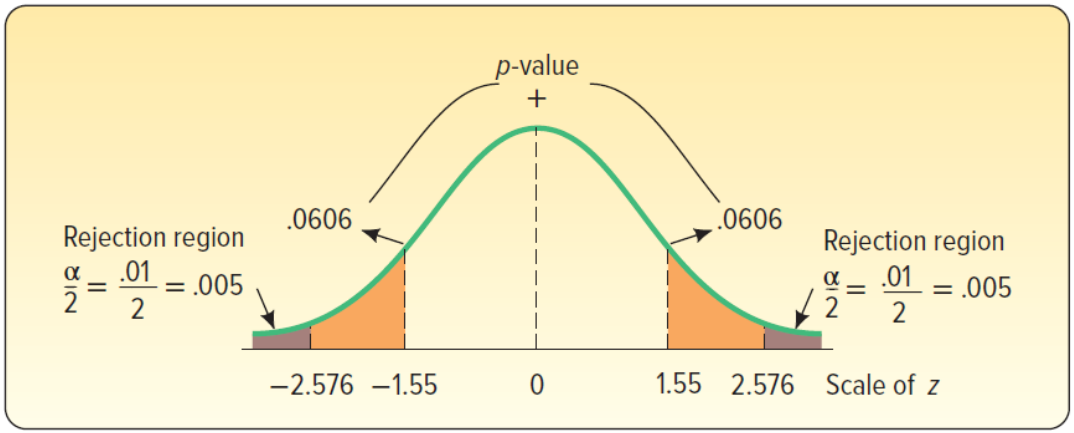
Finding a p-Value

- In the previous example about desk production, the computed z was 1.547 and H_0 was not rejected
- Round the computed z-value to two decimal places, 1.55
- Using the z-table, find the probability of finding a z-value of 1.55 or more by .5000 − .4394 = .0606
- Since this is a two-tailed test 2(.0606) = .1212
- In this chart, we can easily compare the p-value with the level of significance



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Hypothesis Testing, σ Unknown

- When testing a hypothesis about a population mean

TESTING A MEAN, σ UNKNOWN

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \qquad (10\text{–}2)$$

- The major characteristics of the t distribution are
  - It is a continuous distribution
  - It is bell-shaped and symmetrical
  - There is a family of t distributions, depending on the number of degrees of freedom
  - It is flatter, or more spread out, than the standard normal distribution

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Hypothesis Testing, σ Unknown Example

The Myrtle Beach International Airport provides a cell phone parking lot where people can wait for a message to pick up arriving passengers. To decide if the cell phone lot has enough parking places, the manager of airport parking needs to know if the mean time in the lot is more than 15 minutes. A sample of 12 recent customers showed they were in the lot the following lengths of time, in minutes (see below).

At the .05 significance level, is it reasonable to conclude that the mean time in the lot is more than 15 minutes?

30	24	28	22	14	2	39	23	23	28	12	31

Step 1: State the null hypothesis and the alternate hypothesis

$$H0: \mu \leq 15$$
$$H1: \mu > 15$$

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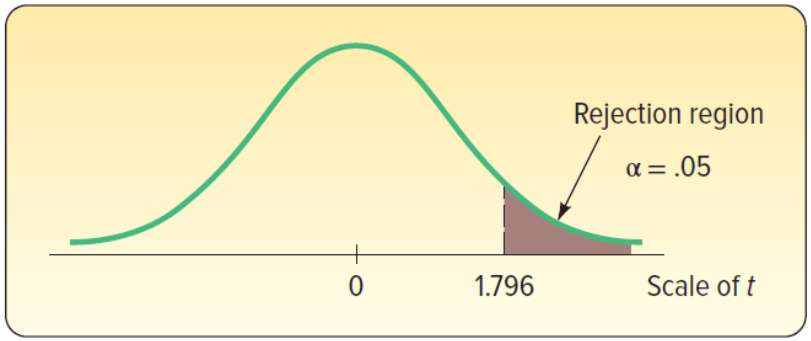
Hypothesis Testing, Unknown Example (2 of 3)

Step 2: Select the level of significance; we will use .05
Step 3: Select the test statistic; we will use t
Step 4: Formulate the decision rule; reject H_0 if t is less than 1.796

	Confidence Intervals				
	80%	90%	95%	98%	99%
	Level of Significance for One-Tailed Test				
df	0.10	0.05	0.025	0.010	0.005
	Level of Significance for Two-Tailed Test				
	0.20	0.10	0.05	0.02	0.01
1	3.078	6.314	12.706	31.321	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	$\boxed{1.796}$	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055



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Hypothesis Testing, Unknown Example (3 of 3)

Step 5: Take sample, make decision

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{23-15}{9.835/\sqrt{12}} = 2.818$$

The test statistic of 8.818 is greater than our critical value of 1.796. Therefore, our decision is: Do not reject H_0

Step 6: Interpret the result; The test results do not allow the claims manager to conclude the cost-cutting measures have been effective.

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Type I and Type II Errors

- Type I error occurs when a true null hypothesis is rejected
  - The probability of making a Type I error is equal to the level of significance, α
- A Type II error occurs when a false null hypothesis is not rejected
  - The probability of making a Type II error is designated, β
  - The likelihood of a Type II error must be calculated comparing the hypothesized distribution to an alternate distribution based on sample results and can be calculated with this formula

TYPE II ERROR

$$z = \frac{\bar{x}_c - \mu_1}{\sigma/\sqrt{n}} \qquad (10\text{–}3)$$

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Type II Error Example

Western Wire Products purchases steel bars to make cotter pins. Past experience indicates that the mean tensile strength of all incoming shipments is 10,000 psi and that the standard deviation is 400 psi. To monitor the quality of the cotter pins, samples of 100 pins are randomly selected and tested for their strength. In our hypothesis testing procedure the hypotheses are:

$$H_0: \mu = 10{,}000$$
$$H_1: \mu \neq 10{,}000$$

Using a 0.05 significance level, accept the shipment if the sample mean strength falls between the critical values 9.922 psi and 10.078 psi. If the sample mean does not fall between the critical values, we conclude the shipment does not meet the quality standard.

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Type II Error Example (2 of 2)

The sample mean, 9.900 psi, is not within the specified range. To calculate the probability of a Type II error, assume the sample mean is the true mean (see graph B). Determine the probability of the sample mean falling between 9.900 and 9.922. Then subtract this probability from .5000 to arrive at the probability of making a Type II error, .2912

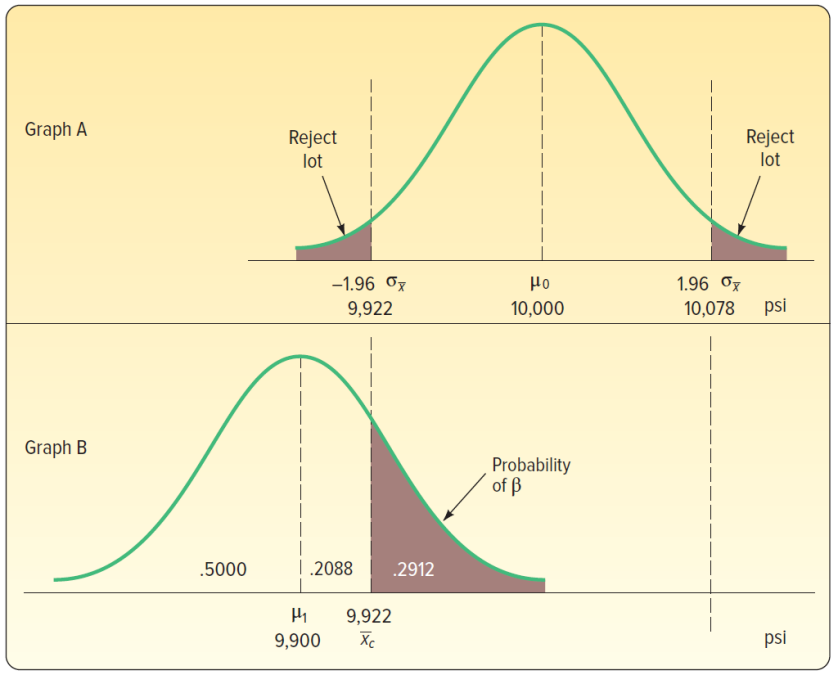

CHART 10–8 Charts Showing Type I and Type II Errors

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Chapter 10 Practice Problems

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Question 7

LO10-2,3,4

A recent national survey found that high school students watched an average (mean) of 6.8 movies per month with a population standard deviation of 1.8. The distribution of number of movies watched per month follows the normal distribution. A random sample of 36 college students revealed that the mean number of movies watched last month was 6.2. At the .05 significance level, can we conclude that college students watch fewer movies a month than high school students?

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Question 13

LO10-2,3,6

The mean income per person in the United States is $60,000, and the distribution of incomes follows a normal distribution. A random sample of 10 residents of Wilmington, Delaware, had a mean of $70,000 with a standard deviation of $10,000. At the .05 level of significance, is that enough evidence to conclude that residents of Wilmington, Delaware, have more income than the national average?

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Question 19

LO10-2,3,5,6

A Washington, D.C., "think tank" announces the typical teenager sent 67 text messages per day in 2017. To update that estimate, you phone a sample of 12 teenagers and ask them how many text messages they sent the previous day. Their responses were:

51	175	47	49	44	54	145	203	21	59	42	100

At the .05 level, can you conclude that the mean number is greater than 67? Compute the *p*-value and describe what it tells you.

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Question 23

LO10-27

The management of KSmall Industries is considering a new method of assembling a computer. The current assembling method requires a mean time of 60 minutes with a standard deviation of 2.7 minutes. Using the new method, the mean assembly time for a random sample of 24 computers was 58 minutes.

Using the .10 level of significance, can we conclude that the assembly time using the new method is faster?

What is the probability of a Type II error?

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